

AN EMPLOYEE REQUESTS INSTALLATION OF LICENSED SOFTWARE THROUGH THE SERVICE CATALOG

Complete Project Documentation

ServiceNow – Service Portal, Service Catalog, Flow Designer, Approvals, ACLs and Notifications

Project	An Employee Requests Installation of Licensed Software through the Service Catalog
Platform	ServiceNow
Primary Interface	Service Portal
Workflow	Catalog → Validation → License Check → Approval → Installation Task → Status → Notification

1. Project Overview

The project focuses on developing an automated Licensed Software Installation Request Management system using ServiceNow. The solution provides employees with a simple and structured way to request installation of approved licensed software through the Service Catalog.

The main purpose is to capture software requirements accurately, validate the request, check license availability, obtain the required approval, create a fulfillment task for the IT/software team, and keep the requester informed through status updates and notifications.

The system uses Service Catalog, Catalog Item, Catalog Variables, Catalog UI Policies, Catalog Client Scripts, Flow Designer, Approval Requests, fulfillment tasks, Access Controls, Notifications, and reference data to manage the software request lifecycle.

Project Workflow

The implemented process follows this sequence:

1. Employee opens the Licensed Software Request catalog item through the Service Portal.
2. Employee enters software name, version, business justification, device details, installation location, and other required information.
3. Catalog validation checks mandatory fields and applies dynamic form behavior.
4. Employee submits the request using Order Now.
5. ServiceNow creates the Request (REQ) and Requested Item (RITM) records.
6. Flow Designer validates the request and checks license availability.
7. The request is routed through the configured approval process.
8. Once approved, the workflow creates a Software Installation Task for the IT/software fulfillment team.
9. Notifications are generated at important stages.

10. The fulfillment team processes the task and updates its status.
11. The completed request can be tracked through ServiceNow.

Project Phases

The project was implemented through the following phases:

- 1. Requirement Analysis & Planning** – Identifying the need for employees to request licensed software through the Service Catalog and defining the required software, user, device, business justification, approval, and installation details.
- 2. Service Catalog & Form Configuration** – Creating the Licensed Software Request Catalog Item, configuring catalog variables and variable sets, and setting mandatory fields and appropriate software choices.
- 3. UI/UX Development & Customization** – Designing the Service Portal request form and configuring Catalog UI Policies and Catalog Client Scripts to control field visibility, mandatory fields, and dynamic form behavior.
- 4. Workflow Automation & Approval** – Using Flow Designer to automate request validation, license availability checking, approval processing, Software Installation Task creation, status updates, and notifications.
- 5. Testing & Security** – Configuring Access Control Lists (ACLs), validating user permissions, performing functional testing, and conducting User Acceptance Testing (UAT) to ensure the complete software request workflow works correctly.
- 6. Deployment, Documentation & Final Presentation** – Preparing the solution for deployment, validating the final configuration, documenting the ServiceNow implementation and testing results, demonstrating the complete licensed software request workflow, and identifying future enhancements.

Final Outcome

The completed project provides a **structured and automated process for employees to request installation of licensed software through the ServiceNow Service Catalog.**

The main outcome is an end-to-end workflow in which the employee submits the software request, provides the required details, and the system validates the information and checks license availability. The request is then routed for approval, and once approved, a **Software Installation Task** is automatically created and assigned to the IT/software fulfillment team.

The system also provides **request status tracking and notifications** throughout the process, giving employees visibility into the progress of their software installation request.

2. Phase 1 – Requirement Analysis & Planning

Step 1: Understand the Project Requirement

The main requirement is to provide employees with a centralized ServiceNow Service Catalog item through which they can request installation of licensed software.

The system should:

- Allow employees to select the required licensed software.
- Capture necessary details such as **Requested For, Software Name, Software Version, Device, Installation Details, and Business Justification.**
- Validate the information entered by the employee.
- Check whether the required software license is available.
- Send the request for the required approval.
- Automatically create an installation task for the IT/software team after approval.
- Allow the employee to track the request status.
- Send notifications at important stages of the request.

Step 2: Identify Stakeholders

The major stakeholders involved in the process are:

- **Employee / Requester** – submits the licensed software installation request.
- **Manager / Approver** – reviews and approves or rejects the request.
- **IT / Software Team** – validates the license and performs the software installation.

- **ServiceNow Administrator** – configures the Service Catalog, workflow, security, and notifications.

Step 3: Define ServiceNow Components

The following ServiceNow components are planned for the project:

- Service Catalog
- Licensed Software Request Catalog Item
- Catalog Variables
- Variable Sets
- Catalog UI Policies
- Catalog Client Scripts
- Flow Designer
- Approval Request
- Software Installation Task
- Notifications
- Access Control Lists (ACLs)
- Request (REQ) and Requested Item (RITM)

Step 4: Define the Request Process

The planned request process is:

Employee → Service Portal → Licensed Software Catalog Item → Enter Details → Submit Request → License Validation → Approval → Installation Task → Software Installation → Status Update → Notification

Step 5: Define Functional Requirements

The system should provide the following functionality:

1. Employee can access the Service Catalog.
2. Employee can open the Licensed Software Request item.
3. Employee can select the required software.
4. Employee can provide the required installation details. 7
5. Mandatory fields must be validated before submission.
6. A request and requested item should be generated after submission. Edx License availability should be checked.
7. The request should be routed for approval.
8. An installation task should be created after approval.
9. The IT team should be able to update the task status.
10. The requester should receive status notifications.
11. The complete request should remain traceable in ServiceNow.

Step 6: Project Planning

The overall project is planned in the following sequence:

Requirement Analysis & Planning → Service Catalog & Form Configuration → UI/UX Customization → Workflow Automation & Approval → Testing & Security → Deployment, Documentation & Final Presentation

3. Phase 2 – Service Catalog & Form Configuration

The second phase of the project focuses on creating and configuring the **Licensed Software Installation Request** in the ServiceNow Service Catalog. This phase converts the requirements identified in Phase 1 into an actual catalog item and request form.

2.1 Create the Service Catalog Item

A dedicated Service Catalog item was created for the software installation request.

Catalog Item: Licensed Software Installation Request

The catalog item acts as the entry point for employees to request software installation. It is designed to collect the required software, employee, device, business, priority, and licensing information before the request is submitted.

The catalog item is connected to the subsequent request and workflow process so that the submitted information can be used for validation, approval, license checking, task creation, and fulfillment.

2.2 Configure Catalog Variables

The request form was configured with variables to capture the information required by the Software Support Team.

The following variables were configured based on the implemented form:

Variable	Purpose
Which software version you want?	Allows the requester to select the required software version.
Which software you want?	Identifies the software to be installed.

Variable	Purpose
Business Justification	Captures the business reason for requesting the software.
By which date you need it?	Captures the required installation date.
Enter employee name	Identifies the employee for whom the software is requested.
Where you want to install your software/device?	Identifies the target device or system.
Enter your manager name	Identifies the employee's manager for approval.
Enter your department name	Captures the department associated with the request.
Urgency	Determines the priority of the software request.
Select License Type	Identifies whether the request requires a licensed or other software entitlement.

The variables ensure that the request contains sufficient information before it enters the automated approval and fulfillment process.

The screenshot displays the configuration interface for a ServiceNow notification. The title bar reads "Notification - Software Request-Catalog Task Assigned". Below this, a blue box contains instructions: "Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify: When to send the notification, Who receives the notification, What content is in the notification." The configuration fields include: Name (Software Request-Catalog Task Assigned), Table (Catalog Task [rc_task]), Mandatory (checkbox), Category (Uncategorized), Application (Global), and Active (checkbox). A section titled "When to send" includes a dropdown menu set to "Record inserted or updated" and a condition "Assignment group is Software Support Team". The page also features a "Preview Notification" button and a "Related Links" section with links to "Show Notification Scripts", "Advanced view", and "Run Point Scan".

Figure 1: Licensed Software Installation Request – Configured Catalog Variables

2.3 Configure the Request Form

The Service Catalog form was designed to make the software request process simple and structured for employees.

The form captures the following major categories of information:

Software Information

- Software name
- Software version
- License type

Requester Information

- Employee name
- Manager name
- Department

Installation Information

- Target device/system
- Required installation date

Business Information

- Business justification

Priority Information

- Urgency

This information is later consumed by the workflow to determine the appropriate approval and fulfillment path.

2.4 Form Validation and Mandatory Fields

Required fields were configured to ensure that employees cannot submit an incomplete software request.

Mandatory information includes the key details required to process the request, such as:

- Software selection

- Software version
- Business justification
- Required date
- Employee name
- Installation device
- Manager
- Department
- Urgency
- License type

This validation improves data quality and prevents incomplete requests from entering the approval and fulfillment process.

2.5 Configure User-Friendly Form Behavior

The Service Catalog form was configured to provide a structured and user-friendly experience.

The form configuration supports:

- Selection of predefined software options.
- Selection of supported software versions.
- Selection of urgency.
- Selection of license type.
- Entry of business justification.
- Selection of the employee/requested-for information.
- Identification of the installation device.
- Capture of manager and department information.
- Required-date selection.

The collected information is carried forward to the **Requested Item (RITM)** and subsequently becomes available to the automated workflow.

2.6 Request and Requested Item Generation

When the employee submits the catalog form, ServiceNow generates the corresponding request records.

The process follows:

Employee → Service Catalog Item → Catalog Variables → Submit → Request (REQ) → Requested Item (RITM)

The RITM contains the submitted catalog variables and becomes the primary record used by the software installation workflow.

The workflow then uses this information for:

- Priority determination
- License availability checking
- Approval processing
- Software asset/license approval
- Catalog task creation
- Fulfillment
- Status updates
- Notifications

The screenshot shows the configuration page for a notification named 'Software Request-Catalog Task Assigned'. The page is divided into two main sections. The top section contains general settings: Name (Software Request-Catalog Task Assigned), Application (Global), Table (Catalog Task [c_task]), Mandatory (unchecked), Category (Uncategorized), and Active (checked). Below this is a note about digests and an 'Allow Digest' checkbox. The bottom section is titled 'When to send' and contains instructions on when notifications can be sent. It includes a 'Send when' dropdown set to 'Record inserted or updated', an 'Inserted' checkbox, and a 'Conditions' section with a filter: 'Assignment group is Software Support Team'. At the bottom of this section are buttons for 'Preview Notification', 'Update', and 'Delete', along with 'Related Links' for 'Show Notification Scripts', 'Advanced view', and 'Run From Search'.

Figure 2: Generated Requested Item with Software Installation Request Details

2.7 Integration with the Workflow

The configured catalog variables are directly used by the **Software Installation Workflow**.

The workflow retrieves the catalog variables from the submitted request and uses them to control the subsequent processing.

The implemented process is:

Catalog Form → RITM → Get Catalog Variables → Priority Processing → License Lookup → Approval → License Decision → Software Asset Approval → Catalog Task → Fulfillment → Notification

This ensures that the information entered by the employee is not lost and is available throughout the complete request lifecycle.

2.8 Example of Configured Request

An example request was successfully created with values similar to:

- **Software Version:** Latest Supported Version
- **Software:** Adobe Acrobat
- **Business Justification:** For files compression

- **Required Date:** 2026-09-15
- **Employee:** David Loo
- **Device:** ASSET-IBM
- **Manager:** Bud Richman
- **Department:** Development
- **Urgency:** Medium
- **License Type:** LICENSED

This confirms that the Service Catalog form is successfully capturing the information required for downstream processing.

The screenshot displays a Service Catalog form titled 'Install Requested Software'. The form is divided into several sections. At the top, there are fields for 'Number' (SC7490001003), 'Assigned to' (David Loo), 'Configuration item' (ASSET-IBM), 'Active' (checked), 'Approval' (Approved), 'Priority' (None), 'State' (Open), 'Request item' (RTN0001004), and 'Requested for' (System Administrator). Below these is a 'Variables' section containing a list of submitted variables: 'Which software version you want?' (Latest Supported Version), 'Which software you want?' (Adobe Acrobat), 'Business Justification' (For file compression), 'By which date you need it?' (2026-09-15), 'Enter employee name' (David Loo), 'Where you want to install your software/device?' (ASSET-IBM), 'Enter your manager name' (Bud Richman), 'Enter your department name' (Development), 'Urgency' (Medium), and 'Select License Type' (LICENSED). At the bottom, there are fields for 'Short description' (Install Requested Software), 'Description', and 'Work notes'.

Figure 3: Example Software Installation Request with Submitted Variables

Phase 3 – UI/UX Development & Customization

The third phase focuses on improving the **user interface and user experience** of the Licensed Software Installation Request in ServiceNow. The objective is to make the catalog request form simple, clear, and easy for employees to complete while ensuring that the required information is captured correctly.

3.1 UI/UX Design Objectives

The following objectives were considered while customizing the request form:

- Provide a simple and user-friendly software request form.
- Use clear and meaningful field labels.
- Organize the request information logically.

- Make important fields mandatory.
- Reduce unnecessary user input.
- Provide appropriate field types such as choice fields and reference fields.
- Ensure that the form is easy for employees to understand and submit.
- Support the approval and fulfillment teams by presenting complete request information.

3.2 Form Layout and Field Organization

The Licensed Software Installation Request form is organized around the information required to process a software installation request.

Software Information

- **Which software you want?**
- **Which software version you want?**
- **Select License Type**

Requester Information

- **Enter employee name**
- **Enter your manager name**
- **Enter your department name**

Installation Information

- **Where you want to install your software/device?**
- **By which date you need it?**

Business Information

- **Business Justification**

Priority Information

- **Urgency**

This arrangement helps the requester provide information in a logical sequence.

3.3 Catalog UI Policies

Catalog UI Policies are used to control the behavior and visibility of catalog variables based on request conditions.

The UI policies help to:

- Display or hide fields when required.
- Make fields mandatory when applicable.
- Make fields read-only when necessary.
- Improve the overall form experience.
- Prevent users from entering unnecessary information.

These policies ensure that the form dynamically responds to the user's selections.

3.4 Catalog Client Scripts

Catalog Client Scripts are used to provide client-side behavior and validation on the software request form.

The customization supports:

- Validation of user-entered information.
- Automatic handling of selected values.
- Preventing incomplete or invalid submissions.
- Improving interaction between catalog variables.
- Providing a smoother request submission experience.

3.5 Mandatory Field Validation

Important fields are configured as mandatory so that the request contains sufficient information before submission.

For example, the requester must provide:

- Software
- Software version
- Business justification
- Required date
- Employee
- Device
- Manager
- Department
- Urgency
- License type

This prevents incomplete requests from entering the workflow.

3.6 User-Friendly Field Configuration

Appropriate variable types are used according to the information being captured.

For example:

Field	UI Purpose
Software	Select the required software
Software Version	Select supported version
Business Justification	Enter business requirement
Required Date	Specify installation date
Employee	Identify requester/recipient
Device	Identify target device
Manager	Identify approver
Department	Identify business unit
Urgency	Select request priority
License Type	Identify licensing category

3.7 Example of Customized Request Form

A completed request demonstrates how the customized UI captures the information required for processing.

Example values include:

- **Software:** Adobe Acrobat
- **Software Version:** Latest Supported Version
- **Business Justification:** For files compression
- **Required Date:** 2026-09-15
- **Employee:** David Loo
- **Device:** ASSET-IBM
- **Manager:** Bud Richman
- **Department:** Development

- **Urgency:** Medium
- **License Type:** LICENSED

The screenshot shows a web-based form titled 'Catalog Task' with the identifier 'SCM00000000'. The form is divided into several sections. At the top, there are fields for 'Number' (SCM00000000), 'Assigned to', 'Configuration Item', 'Approval' (Approved), 'Priority' (None), 'State' (Open), 'Request item' (RITM00000000), and 'Requested for' (System Administrator). Below these is a 'Variables' section containing several dropdown menus and text input fields. The variables include: 'Which software version you want?' (Latest Supported Version), 'Which software you want?' (Adobe Acrobat), 'Business Justification' (For file compression), 'By which date you need it?' (2024-09-12), 'Enter employee name' (David Lee), 'Where you want to install your software/device?' (RUST-004), 'Enter your manager name' (Bull Richman), 'Enter your department name' (Development), 'Urgency' (Medium), and 'Select License Type' (LICENSED). At the bottom, there are fields for 'Short description' (Install Requested Software), 'Description', and 'Work notes'.

Figure 3.1: Licensed Software Installation Request – User-Friendly Catalog Form with Configured Fields

3.8 UI/UX and Workflow Integration

The customized form is designed to work with the subsequent automation process:

User Interface → Catalog Variables → Request Submission → RITM → License Validation → Approval → Installation Task → Fulfillment

Therefore, the UI/UX configuration is not only focused on appearance but also ensures that the information collected from the employee can be consumed by the workflow.

3.9 Phase 3 Outcome

At the end of Phase 3:

- The catalog form has a clear and structured layout.
- Catalog variables are presented in a user-friendly manner.
- Mandatory-field validation is implemented.
- UI Policies and Client Scripts support dynamic form behavior.
- The form captures all information required for approval and fulfillment.
- The customized form is ready for integration with the automated workflow.

Phase 4 – Workflow Automation & Approval

The fourth phase focuses on automating the complete processing of the **Licensed Software Installation Request** after the employee submits the Service Catalog form. The objective is to ensure that requests are validated, license availability is checked, approvals are obtained, installation tasks are created, and users are notified automatically.

4.1 Workflow Automation Overview

The workflow is designed to reduce manual processing and ensure that every software request follows a controlled process.

Employee → Catalog Request → RITM → Validation → License Check → Approval → License Decision → Software Asset Approval → Catalog Task → Fulfillment → Notification

The workflow is implemented using the ServiceNow workflow/Flow Designer environment.

4.2 Workflow Configuration

The configured workflow is:

SIRA_Software_Installation_Workflow

The workflow processes the Requested Item (RITM) generated from the **Licensed Software Installation Request** catalog item.

The workflow contains activities for:

- Retrieving catalog variables
- Processing request priority
- Checking software license availability
- Requesting approval
- Processing the approval decision
- Performing software asset/license approval
- Creating the catalog task
- Assigning the task to the **Software Support Team**
- Completing the fulfillment process
- Sending notifications

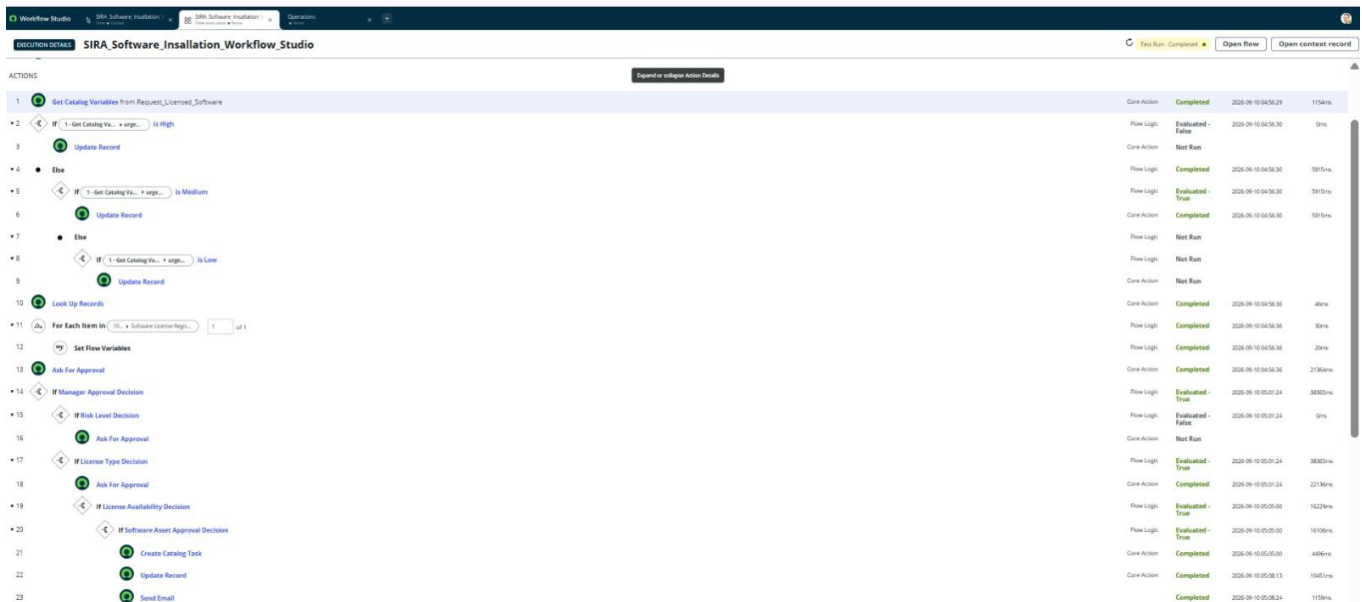


Figure 4.1: SIRA Software Installation Workflow – Workflow Automation Configuration

4.3 Get Catalog Variables

After the employee submits the request, the workflow retrieves the catalog variables from the Requested Item.

The variables include:

- Software
- Software Version
- Business Justification
- Required Date
- Employee
- Device
- Manager
- Department
- Urgency
- License Type

These values are then used by the workflow for validation, priority processing, license checking, approval, and fulfillment.

4.4 Priority Processing

The workflow processes the request based on the selected **Urgency**.

The priority information is used to determine the appropriate SLA and processing path.
For example:

- **High urgency** → High-priority SLA
- **Medium urgency** → Normal-priority SLA
- **Low urgency** → Low-priority processing

This ensures that urgent software installation requests receive appropriate attention.

4.5 License Availability Check

The workflow checks whether a valid license is available for the requested software.

The basic logic is:

Requested Software → Check License Inventory → License Available?

If the license is available, the request proceeds toward approval and fulfillment.

If the license is unavailable, the request can be routed through the appropriate decision or exception path rather than proceeding directly to installation.

4.6 Approval Process

After the initial validation and license check, the request is routed for approval.

The manager/appropriate approver reviews the request based on:

- Requested software
- Business justification
- Employee
- Department
- Device
- Required date
- Urgency
- License information

The approval decision determines whether the request can proceed.

Notification - Software Request-Approval Required

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify:

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Application:

* Table: Active: ☒

Mandatory: ☐

* Category:

A digest combines and summarizes multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent if the specified Conditions are met under one of the following circumstances:

- A record is inserted or updated into the Table specified above.
- The specified event is fired
- Via a Flow Action

Send when: Updated: ☐

Inserted: ☒

Conditions:

State is

Related Links

[Show Notification Scripts](#)

[Advanced user](#)

[Run Report Script](#)

[ITN Utility Versions](#)

Figure 4.2: Software Request – Approval Required Notification Configuration

4.7 Approval Decision

The workflow evaluates the approval result.

If Approved

The workflow continues to the next stages:

Approval → Software Asset Approval → Catalog Task → Software Installation

If Rejected

The request does not proceed to software installation and the requester is informed about the approval decision.

Notification - Software Request-Approval Decision

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name

Software Request-Approval Decision

Application

Global

Active

Table

Requested Item [sc_req_item]

Mandatory

Category

Uncategorized

A digest combines and summarizes multiple updates for a single or multiple target records within a set time interval.

Allow Digest

When to send

Who will receive

What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when

Record inserted or updated

Updated

Inserted

Conditions

Add Filter Condition

Add OR Clause

All of these conditions must be met

State

is

Closed Complete

AND

OR

Approval

is

Approved

AND

OR

OR all of these conditions must be met

State

is

Closed Incomplete

AND

OR

Approval

is

Rejected

AND

OR

Preview Notification

Update

Delete

Related Links

Show Notification Scripts
Advanced view
Run Point Scan

Figure 4.3: Software Request – Approval Decision Notification

4.8 Software Asset Approval

After the required request approval, the workflow performs the software asset/license approval step.

This ensures that the software installation is consistent with the organization's licensing requirements and available software inventory.

The process helps prevent unauthorized or unlicensed software installations.

4.9 Catalog Task Creation

Once the request passes the required approval and license checks, the workflow creates a **Catalog Task (sc_task)** for the software support team.

The task contains the information required to perform the installation.

The task is assigned to:

Software Support Team

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The support team can then process the installation and update the task status.

The screenshot shows the 'Notification - Software Request-Fulfillment Completed' configuration page in ServiceNow. The page has a dark blue header with the notification name. Below the header, there's a light blue box with instructions: 'Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify: When to send the notification, Who receives the notification, What content is in the notification.' The main configuration area includes fields for Name (Software Request-Fulfillment Completed), Application (Global), Table (Catalog Task [bc_task]), Mandatory (checkbox), Category (Uncategorized), and Active (checkbox). Below these, there's a section for 'When to send' with tabs for 'When to send', 'Who will receive', and 'What it will contain'. The 'When to send' tab is active, showing a list of circumstances: 'A record is Inserted or Updated into the Table specified above', 'The specified event is fired', and 'Via a Flow Action'. The 'Send when' dropdown is set to 'Record inserted or updated', and the 'Updated' checkbox is checked. The 'Conditions' section shows 'State' is 'Closed Complete' with 'AND' logic. At the bottom, there are buttons for 'Preview Notification', 'Update', and 'Delete', and a 'Related Links' section with links to 'Show Notification Scripts', 'Advanced view', 'Run Point Scan', and 'ISN Utilis Versions (1)'.

Figure 4.4: Software Request – Catalog Task Assigned Notification Configuration

4.10 Fulfillment Process

The fulfillment process follows:

Approved Request → Catalog Task → Software Support Team → Software Installation → Task Completion

The support team installs the requested software on the specified device and updates the catalog task.

Once fulfillment is completed, the request can move toward closure.

4.11 Notifications

Automated notifications keep the requester and support teams informed throughout the process.

The configured notifications include:

1. **Software Request – Approval Required**
2. **Approval Record Assigned – Request**

3. Software Request – Approval Decision
4. Software Request – Catalog Task Assigned
5. Software Request – Fulfillment Completed

These notifications provide visibility into important request events.

4.12 SLA and Priority Automation

SLAs are configured to monitor request processing time.

Examples include:

SLA	Target
Software_Request_High_Priority	1 Day
Software_Request_Normal_Priority	2 Days

The SLA conditions are associated with the request state and urgency so that the system can monitor whether requests are completed within the expected time.

The screenshot displays the 'Software Request_Normal Priority' SLA configuration. At the top, a status bar indicates the SLA is active and will breach on 2026-09-17 11:00:00. The configuration form includes fields for Name, Type (SLA), Target (None), Table (Requested item [rc_req_item]), and Flow (Default SLA flow). The Application is set to Global, and the Duration type is User specified duration. The Duration is set to 2 Days, 00 Hours, 00 Minutes, and 00 Seconds. The Schedule source is SLA definition, and the Schedule is set to 8-5 weekdays excluding holidays. The Timezone source is The caller's time zone. Below the form, there are buttons for Start condition, Pause condition, Stop condition, and Reset condition. The Start condition section shows a filter condition: State is Open AND Urgency is 2 - Medium. The Cancel condition section is empty. At the bottom, there are links for Related Links, Show Schedule, Update SLA Definition, and SLA Link Versions (2). The bottom section shows a table for SLA Breakdown definitions, which is currently empty.

Figure 4.5: Software Request SLA Configuration – High Priority

Name: Software Request_High Priority

Type: SLA

Target: -- None --

Table: Requested Item [rc_req_item]

Flow: Default SLA flow

Enable logging: ☐

Active: ☒

Application: Global

Duration type: User specified duration

* Duration: Days 1 Hours 00 00 00

Schedule source: SLA definition

* Schedule: 8-5 weekdays excluding holidays

Timezone source: The caller's time zone

Start condition: Add Filter Condition Add OR Clause

All of these conditions must be met

State is Open

Urgency is 1 - High

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition Add OR Clause

-- choose field -- -- oper -- -- value --

Update Delete

Related Links

Show Schedule

Validate SLA Definition

SN Utilist Versions (2)

SLA Breakdown definitions

SLA Definition - Software Request_High Priority

SLA breakdown definition

No records to display

Figure 4.6: Software Request SLA Configuration – Normal Priority

4.13 Update Set

The workflow automation configuration is captured in the update set:

SIRA_Workflow_Automation

This allows the workflow-related configuration to be tracked and moved between ServiceNow instances as required.

4.14 End-to-End Workflow

The complete automated process is:

Employee



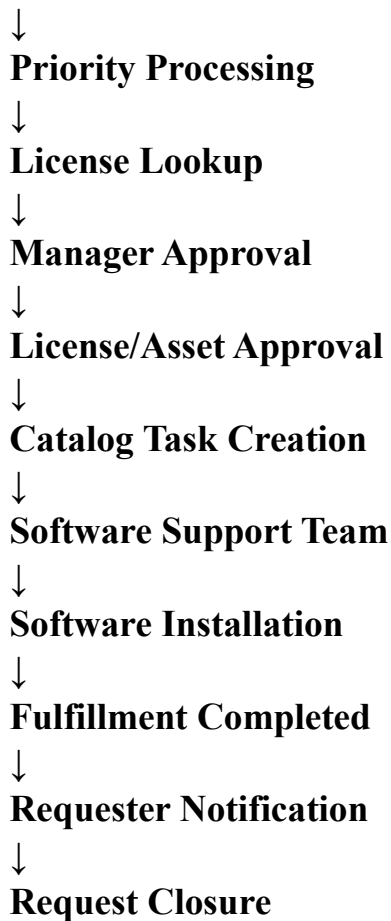
Licensed Software Installation Request



Request (REQ) / Requested Item (RITM)



Get Catalog Variables



4.15 Phase 4 Outcome

At the end of Phase 4:

- The software request workflow is automated.
- Catalog variables are retrieved automatically.
- Request priority is processed.
- License availability is checked.
- Approval is automated.
- Approval decisions control the next workflow step.
- Catalog tasks are automatically created.
- Tasks are assigned to the **Software Support Team**.
- SLAs monitor request processing time.
- Notifications keep users informed.
- The complete request lifecycle is traceable.

Phase 5 – Testing & Security

The fifth phase focuses on validating the **Licensed Software Installation Request** to ensure that the configured catalog form, workflow, approvals, notifications, SLAs, and security controls work correctly. Testing is performed from the employee, approver, and IT support perspectives.

5.1 Testing Objectives

The main objectives of this phase are:

- Verify that the catalog request form works correctly.
- Confirm that mandatory fields and validations work as expected.
- Verify REQ and RITM creation.
- Validate license availability processing.
- Verify approval routing and approval decisions.
- Confirm automatic Catalog Task creation.
- Verify assignment to the **Software Support Team**.
- Validate SLA behavior based on urgency.
- Verify all configured notifications.
- Confirm request status updates and closure.
- Ensure appropriate access and security controls are applied.

5.2 Functional Testing

Functional testing verifies that each major function of the software request process operates according to the requirements.

Test Area	Expected Result
Catalog item access	Employee can access the software request
Mandatory fields	Request cannot be submitted with required data missing
Software selection	Selected software is captured correctly
Request submission	REQ and RITM are created
License validation	License availability is checked
Approval	Request is routed to the appropriate approver

Test Area	Expected Result
Approval decision	Approved/rejected paths work correctly
Catalog Task	Task is created after required approval
Assignment	Task is assigned to Software Support Team
Fulfillment	Installation task can be completed
Notification	Relevant users receive notifications
Closure	Completed request reaches the appropriate final state

5.3 Catalog Form Testing

The catalog form is tested with valid and invalid inputs.

Valid Input Testing

A complete request is submitted with all required values, such as:

- Software: **Adobe Acrobat**
- Software Version: **Latest Supported Version**
- Business Justification: **For files compression**
- Required Date: **2026-09-15**
- Employee: **David Loo**
- Device: **ASSET-IBM**
- Manager: **Bud Richman**
- Department: **Development**
- Urgency: **Medium**
- License Type: **LICENSED**

The system should successfully submit the request and create the corresponding RITM.

Invalid Input Testing

The form is also tested by:

- Leaving mandatory fields empty.
- Entering incomplete information.
- Selecting invalid or unsupported values.

- Attempting to submit without required request details.

The system should prevent incomplete requests from being submitted.

5.4 Workflow Testing

The complete workflow is tested from submission through fulfillment.

Employee Submission → RITM → License Check → Approval → Asset Approval → Catalog Task → Fulfillment → Notification → Closure

Each workflow stage is verified to ensure that the request moves to the correct next step.

5.5 Approval Testing

Approval testing verifies both possible approval outcomes.

Approved Request

Request → Approval → Approved → Catalog Task → Software Installation

The approved request should continue toward fulfillment.

Rejected Request

Request → Approval → Rejected → Requester Notification

A rejected request should not proceed to software installation.

5.6 Notification Testing

The following notifications are tested:

1. **Software Request – Approval Required**
2. **Approval Record Assigned – Request**
3. **Software Request – Approval Decision**
4. **Software Request – Catalog Task Assigned**
5. **Software Request – Fulfillment Completed**

The notification configuration and delivery are verified to ensure that the correct recipient receives the appropriate message at the correct stage.

Notification - Software Request-Approval Required

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Application:

* Table: Active: ☒

Mandatory: ☐

* Category:

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Send when: Updated: ☐

Inserted: ☒

Conditions:

State: is

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Print Scan](#)

[\[SN Utility\] Versions \(2\)](#)

Figure 5.1: Software Request – Approval Required Notification

Notification - Software Request-Fulfillment Completed

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Application:

* Table: Active: ☒

Mandatory: ☐

* Category:

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Send when: Updated: ☒

Inserted: ☐

Conditions:

State: is

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Print Scan](#)

[\[SN Utility\] Versions \(3\)](#)

Figure 5.2: Software Request – Catalog Task Assigned Notification

5.7 Email Validation

Email records are reviewed to verify that notifications were generated successfully.

Testing includes:

- Notification name
- Recipient
- Subject
- Message content
- Trigger condition
- Date/time generated
- Request/RITM reference

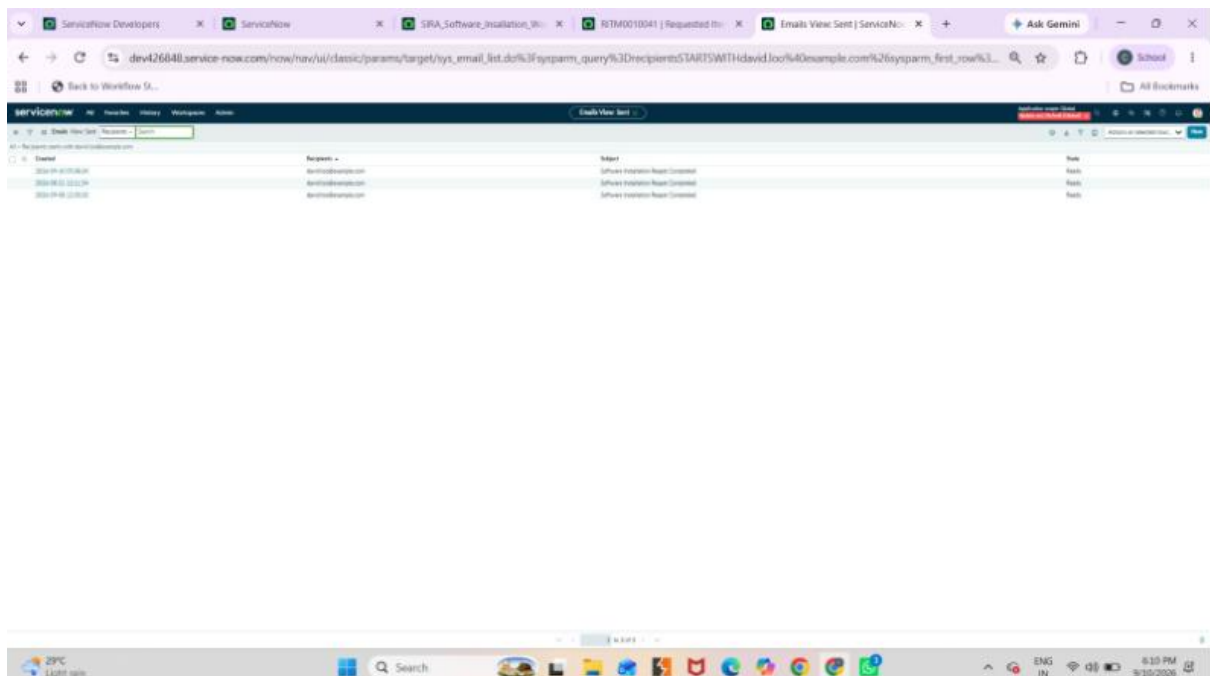


Figure 5.3: Software Request Notification – Sent Email Record

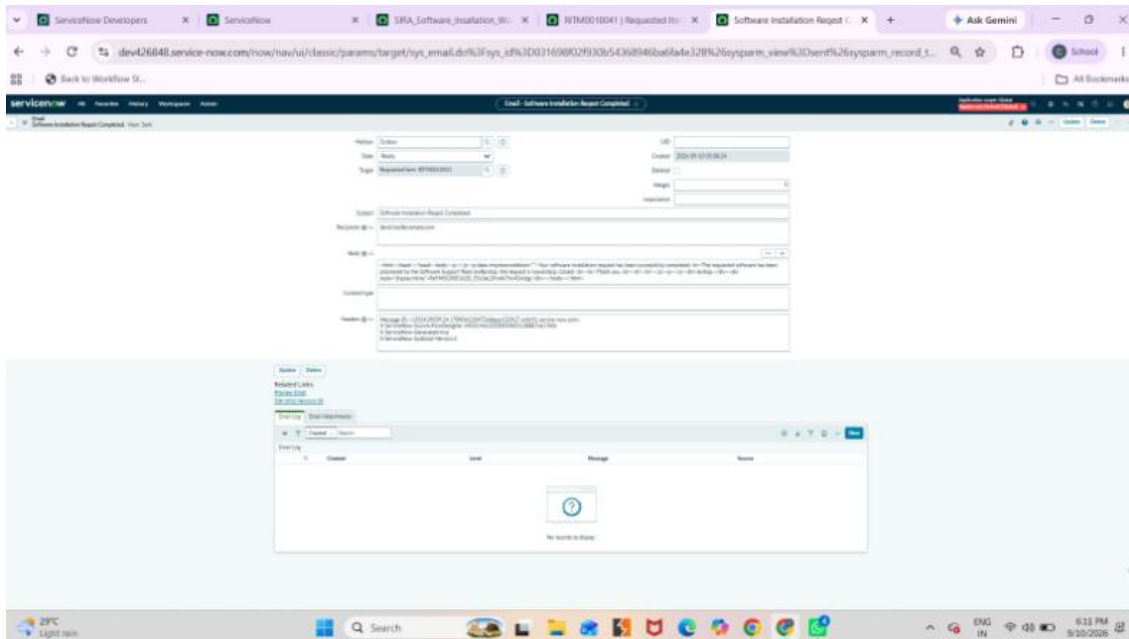


Figure 5.4: Software Request Notification – Email Preview

5.8 SLA Testing

SLA testing verifies that the request receives the appropriate SLA based on urgency.

Configured examples include:

Priority SLA Target

High 1 Day

Medium 2 Days

Low Based on configured low-priority SLA

The SLA should start when its defined conditions are satisfied and should monitor the request until completion or the applicable stop condition.

An SLA starting now will breach on 2026-09-17 11:00:00 (Actual elapsed time: 7 Days 6 Hours 23 Minutes)

Name: Software Request_Normal Priority	Application: Global
Type: SLA	Duration type: User specified duration
Target: -- None --	* Duration: Days 2 Hours 00 00 00
Table: Requested item [it_req_item]	Schedule source: SLA definition
Flow: Default SLA flow	* Schedule: 8-5 weekdays excluding holidays
Enable logging: ☐	Timezone source: The caller's time zone
Active: ☒	

Start condition | Pause condition | Stop condition | Reset condition

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition | Add OR Clause

All of these conditions must be met

State	is	Open	AND	OR	X
Urgency	is	2 - Medium	AND	OR	X

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition | Add OR Clause

-- choose field -- | -- op -- | -- value --

Update | Delete

Related Links
[Show Schedule](#)
[Validate SLA Definition](#)
[ISN Util Versions](#)

SLA Breakdown definitions | SLA breakdown def | Search

SLA Definition = Software Request_Normal Priority

SLA breakdown definition

No records to display

Figure 5.5: Software Request – High Priority SLA Configuration

Name: Software Request_High Priority	Application: Global
Type: SLA	Duration type: User specified duration
Target: -- None --	* Duration: Days 1 Hours 00 00 00
Table: Requested item [it_req_item]	Schedule source: SLA definition
Flow: Default SLA flow	* Schedule: 8-5 weekdays excluding holidays
Enable logging: ☐	Timezone source: The caller's time zone
Active: ☒	

Start condition | Pause condition | Stop condition | Reset condition

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition | Add OR Clause

All of these conditions must be met

State	is	Open	AND	OR	X
Urgency	is	1 - High	AND	OR	X

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition | Add OR Clause

-- choose field -- | -- op -- | -- value --

Update | Delete

Related Links
[Show Schedule](#)
[Validate SLA Definition](#)
[ISN Util Versions](#)

SLA Breakdown definitions | SLA breakdown def | Search

SLA Definition = Software Request_High Priority

SLA breakdown definition

No records to display

Figure 5.6: Software Request – Normal Priority SLA Configuration

5.9 Security Testing

Security testing ensures that only authorized users can access or modify appropriate information.

The following areas are considered:

- User roles
- Catalog item access
- Requester access
- Approver access
- IT support access
- ACL restrictions
- Catalog Task access
- Sensitive request information
- Unauthorized modification prevention

The objective is to ensure that employees can submit requests while administrative and fulfillment functions remain restricted to authorized personnel.

5.10 Role and Access Validation

Access is tested from different user perspectives.

User	Expected Access
Employee	Submit and view own software request
Manager/Approver	Review and approve assigned requests
Software Support Team	Access and process assigned Catalog Tasks
ServiceNow Administrator	Configure and administer the complete process

This ensures separation of responsibilities between requesters, approvers, support teams, and administrators.

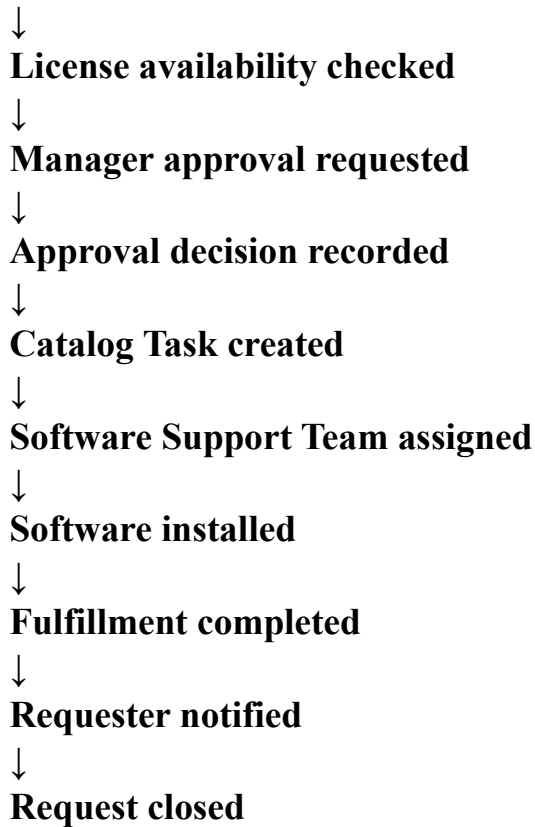
5.11 End-to-End Testing

The complete scenario is tested using a real software installation request.

Employee submits request

↓

REQ/RITM created



Successful completion of this scenario confirms that the major components work together correctly.

5.12 Testing Results

The testing phase confirms that:

- The catalog form accepts valid requests.
- Mandatory fields are enforced.
- REQ/RITM records are generated correctly.
- Workflow activities execute in the expected sequence.
- Approval routing works correctly.
- License validation is incorporated into the process.
- Catalog Tasks are created and assigned correctly.
- Notifications are generated.
- SLAs are applied according to priority.
- Security and access controls restrict unauthorized actions.

Phase 6 – Deployment, Documentation & Final Presentation

The sixth and final phase focuses on preparing the **Licensed Software Installation Request** application for deployment, documenting the completed implementation, and presenting the final solution. This phase ensures that the configured ServiceNow components are properly packaged, validated, and ready for demonstration or transfer to another environment.

6.1 Deployment Preparation

Before deployment, the completed configuration is reviewed to ensure that all required components are included.

The deployment package includes:

- Service Catalog Item
- Catalog Variables
- Catalog UI Policies
- Catalog Client Scripts
- Workflow / Flow Designer configuration
- Approval configuration
- License validation logic
- Catalog Tasks
- Notifications
- SLAs
- Security and access configuration
- Dashboard configuration

6.2 Update Set Configuration

The project configuration is captured using the ServiceNow Update Set:

SIRA_Workflow_Automation

The update set provides a controlled method for tracking configuration changes and moving the developed solution between ServiceNow instances.

The update set contains the required customer updates associated with the workflow automation and related configuration.

Name	Application	State	Update source	Description	Loaded	Committed	Parent	Remote Batch Base
SIRA_Dashboard_Migration	Global	Committed	(empty)		2026-09-06 07:59:46	2026-09-06 08:02:40	(empty)	(empty)
SIRA_Software_Installation_Request_Autom...	Global	Committed	(empty)		2026-09-06 07:55:42	2026-09-06 07:57:38	(empty)	(empty)
SIRA_Workflow_Automation	Global	Committed	(empty)	Workflow automation for the Software Inc...	2026-09-06 08:56:30	2026-09-06 08:55:05	(empty)	(empty)

related Links
[Export Update Set from SRS](#)

Figure 6.1: SIRA_Workflow_Automation – Update Set Configuration

6.3 Review of Update Set Customer Updates

The Customer Updates section is reviewed to verify that the required configuration changes have been captured.

The review helps confirm that important components such as workflow, notifications, scripts, and other configuration records are included before deployment.



Figure 6.2: SIRA_Workflow_Automation – Captured Customer Updates

6.4 Update Set Completion and Retrieval

After development and testing are completed, the update set is prepared for deployment.

The deployment process includes:

Development → Testing → Update Set Completion → Export/Retrieve → Target Instance → Validation

The retrieved update set is reviewed before committing the changes to the target environment.

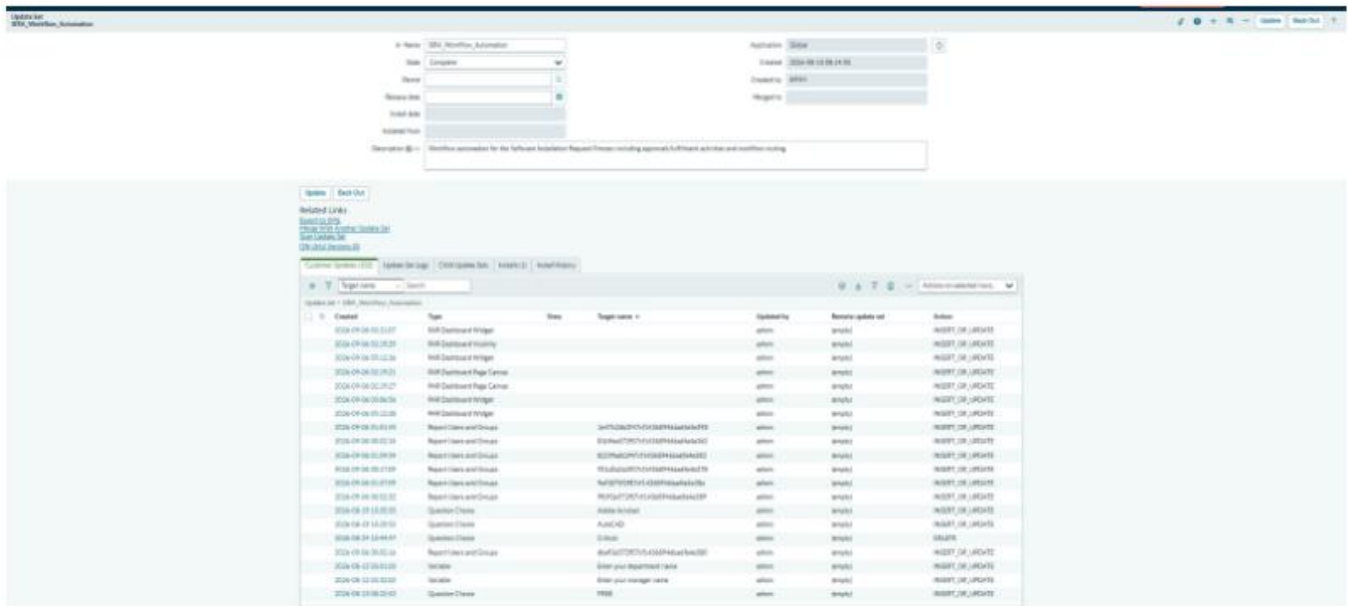


Figure 6.3: SIRA_Workflow_Automation – Retrieved/Committed Update Set

6.5 Post-Deployment Validation

After deployment, the application is validated to ensure that the configuration continues to work correctly.

The following are checked:

- Catalog item availability
- Catalog variables
- Request submission
- REQ/RITM generation
- Workflow execution
- Approval routing
- License validation
- Catalog Task creation
- Assignment to Software Support Team
- Notifications
- SLA processing
- Request completion

This ensures that deployment has not affected the functionality tested during Phase 5.

6.6 Software Request Management Dashboard

A ServiceNow dashboard is configured to provide visibility into software requests and license information.

The dashboard is:

Software Request Management Dashboard

The dashboard provides useful operational information through visual reports and metrics.

The configured dashboard metrics include:

- **Software Requests by State**
- **Software License Inventory by License Type**
- **Software Request SLA Compliance**
- **Software License Availability by Software**

These reports help administrators and support teams monitor request processing, SLA performance, and license availability.

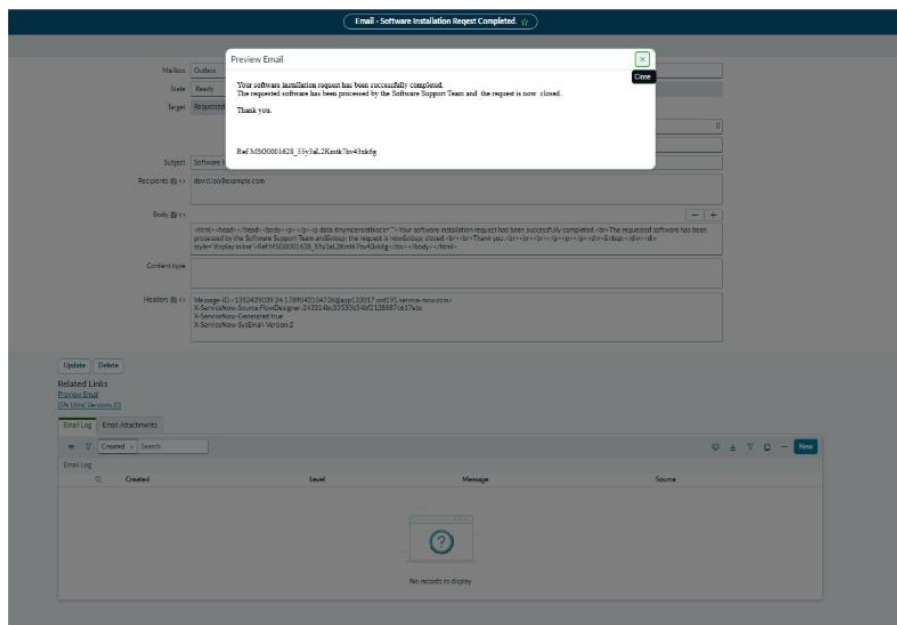


Figure 6.4: Software Request Management Dashboard

6.7 Documentation

The project documentation records the complete implementation process, including:

Phase 1 – Requirement Analysis & Planning

Requirement identification, stakeholders, functional requirements, and project planning.

Phase 2 – Service Catalog & Form Configuration

Catalog item, variables, form configuration, validation, and REQ/RITM generation.

Phase 3 – UI/UX Development & Customization

Form layout, UI Policies, Client Scripts, mandatory fields, and user experience improvements.

Phase 4 – Workflow Automation & Approval

Workflow, license validation, approval process, catalog task creation, notifications, and SLAs.

Phase 5 – Testing & Security

Functional testing, workflow testing, approval testing, notification testing, SLA testing, security testing, and end-to-end validation.

Phase 6 – Deployment, Documentation & Final Presentation

Update Set, deployment, dashboard, final validation, documentation, and project presentation.

6.8 Final Demonstration Flow

The final presentation demonstrates the complete employee software request lifecycle:

Employee



Service Catalog



Licensed Software Installation Request



REQ / RITM Creation



License Validation



Manager Approval



Software Asset Approval



Catalog Task



Software Support Team



Software Installation



Fulfillment Completed



Requester Notification



Dashboard / Reporting



Request Closure

6.9 Final Project Deliverables

The completed project provides:

1. Configured Licensed Software Installation Request catalog item.
2. Configured catalog variables and form validation.
3. Customized UI/UX.
4. Automated software installation workflow.
5. License availability validation.
6. Approval mechanism.
7. Automated Catalog Task creation.
8. Software Support Team assignment.
9. SLA configuration.
10. Automated notifications.
11. Security and role-based access.
12. Software Request Management Dashboard.
13. Update Set for deployment.
14. Complete project documentation.
15. Final project demonstration.

6.10 Phase 6 Outcome

At the end of Phase 6, the **Licensed Software Installation Request** solution is fully documented, tested, packaged using the appropriate Update Set, and prepared for deployment and final presentation.

The final solution provides an end-to-end automated process for requesting, approving, validating, installing, tracking, and completing licensed software requests in ServiceNow.

Final Workflow – Image Evidence Set

The following screenshots provide visual evidence for the complete process, from catalog request and workflow automation through approval, fulfillment, SLA monitoring, dashboard reporting, and deployment.

The screenshot displays the 'SLA Definition' configuration page in ServiceNow. At the top, a status bar indicates: 'An SLA starting now will breach on 2026-09-17 11:00:00 (Actual elapsed time: 7 Days 6 Hours 23 Minutes)'. The main configuration area is divided into two columns. The left column contains fields for 'Name' (Software Request_Normal Priority), 'Type' (SLA), 'Target' (None), 'Table' (Requested Item [rc_req_item]), 'Flow' (Default SLA flow), 'Enable logging' (unchecked), and 'Active' (checked). The right column contains 'Application' (Global), 'Duration type' (User specified duration), 'Duration' (2 Days, 00 Hours, 00 Minutes), 'Schedule source' (SLA definition), 'Schedule' (8-5 weekdays excluding holidays), and 'Timezone source' (The caller's time zone). Below these fields are tabs for 'Start condition', 'Pause condition', 'Stop condition', and 'Reset condition'. The 'Start condition' tab is active, showing a rule: 'All of these conditions must be met' with two conditions: 'State is Open' and 'Urgency is 2 - Medium'. There are 'AND' and 'OR' buttons to modify the logic. Below the conditions are options for 'Retroactive start' (unchecked), 'When to cancel' (Cancel conditions are met), and a 'Cancel condition' section. At the bottom, there are 'Update' and 'Delete' buttons, a 'Related Links' section with links to 'Show Schedule', 'Validate SLA Definition', and 'SLN Util Versions (2)', and a table titled 'SLA Breakdown definitions' with a search bar and a 'No records to display' message.

Figure 1: Licensed Software Installation Request – Submitted Request / Configured Variables

Name: Software Request_High Priority

Type: SLA

Target: -- None --

Table: Requested Item [rc_req_item]

Flow: Default SLA flow

Enable logging: ☐

Active: ☒

Application: Global

Duration type: User specified duration

* Duration: Days 1 Hours 00 Mins 00 Secs 00

Schedule source: SLA definition

* Schedule: 8-5 weekdays excluding holidays

Timezone source: The caller's time zone

Start condition: Pause condition Stop condition Reset condition

The conditions under which the new SLA will be attached and canceled

Start condition: Add Filter Condition Add OR Clause

All of these conditions must be met

State is Open

Urgency is 1 - High

AND OR

Retroactive start: ☐

When to cancel: Cancel conditions are met

Cancel condition: Add Filter Condition Add OR Clause

-- choose field -- -- opp -- -- value --

Update Delete

Related Links

Show Schedule

Validate SLA Definition

SLA Util Versions (2)

SLA Breakdown definitions: SLA breakdown def

SLA Definition = Software Request_High Priority

SLA breakdown definition

No records to display

Figure 2: Licensed Software Installation Request – Catalog Form Evidence

Notification - Software Request-Approval Required

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Software Request-Approval Required

Application: Global

Active: ☒

* Table: Approval [sysapproval_approver]

Mandatory: ☐

* Category: Uncategorized

A digest combines and summarizes multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send: Who will receive: What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is Inserted or Updated into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated

Updated: ☐

Inserted: ☒

Conditions: Add Filter Condition Add OR Clause

State is Requested

AND OR

Preview Notification Update Delete

Related Links

Show Notification Scripts

Advanced view

Run Point Scan

SLN Util Versions (2)

Figure 3: SIRA Software Installation Workflow – Workflow Automation Configuration

Notification - Approval Record Assigned - Request

ord click here. [SN Utilis] Switch to **Service Catalog for mobile** application click here

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Approval Record Assigned - Request Application: Service Catalog for mobile

Table: Approval [sysapproval_approver] Active: ☒

Mandatory: ☐ Category: Service Catalog

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated Updated: ☒

Inserted: ☒

Conditions: stateCHANGESTorequested=sysapproval.sys_class_name=sc_request^EQ

Related Links

[Advanced view](#)

[Run Point Scan](#)

[\[SN Utilis\] Versions](#)

Figure 4: Software Request – Approval Required Notification Configuration

Notification - Software Request-Approval Decision

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Software Request-Approval Decision Application: Global

* Table: Requested Item [sc_req_item] Active: ☒

Mandatory: ☐ Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the **Table** specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated Updated: ☒

Inserted: ☐

Conditions: Add Filter Condition Add OR Clause

All of these conditions must be met

State	is	Closed Complete	AND	OR	X
Approval	is	Approved	AND	OR	X

OR all of these conditions must be met

State	is	Closed Incomplete	AND	OR	X
Approval	is	Rejected	AND	OR	X

Preview Notification Update Delete

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Point Scan](#)

Figure 5: Approval Record Assigned – Request Notification

Notification - Software Request-Fulfillment Completed

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Software Request-Fulfillment Completed Application: Global

* Table: Catalog Task [sc_task] Active: ☒

Mandatory: ☐

* Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated Updated: ☒

Inserted: ☐

Conditions: Add Filter Condition Add OR Clause

State is Closed Complete AND OR

Preview Notification Update Delete

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Point Scan](#)

[\[SN Utilis\] Versions \(1\)](#)

Figure 6: Software Request – Approval Decision Notification

Notification - Software Request-Fulfillment Completed

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Software Request-Fulfillment Completed Application: Global

* Table: Catalog Task [sc_task] Active: ☒

Mandatory: ☐

* Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is **Inserted** or **Updated** into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated Updated: ☒

Inserted: ☐

Conditions: Add Filter Condition Add OR Clause

State is Closed Complete AND OR

Preview Notification Update Delete

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Point Scan](#)

[\[SN Utilis\] Versions \(1\)](#)

Figure 7: Software Request – Catalog Task Assigned Notification Configuration

Notification - Software Request-Catalog Task Assigned

Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify:

- When to send the notification
- Who receives the notification
- What content is in the notification

Name: Application:

* Table: Active: ☒

Mandatory: ☐

* Category:

A digest combines and summarizes multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent if the specified Conditions are met under one of the following circumstances:

- A record is Inserted or Updated into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when: Updated: ☐

Inserted: ☒

Conditions:

is

Related Links

[Show Notification Scripts](#)

[Advanced view](#)

[Run Script Steps](#)

[JSON Utilis Versions \(2\)](#)

Figure 8: Software Request – Fulfillment Completed Notification

Creating Task: SCREQ000000000

Number: Approved:

Assigned to: Priority:

Configuration item: State:

Active: ☒ Request item: Requested for:

Variables

* Which software version you want?

Latest Available Version

* Which software you want?

Adobe Acrobat

* Business Justification

For File compression:

* Requested date you need it?

2024-09-12

* Enter employee name

David Lee

* Where you want to install your software/device?

XOSBT-0001

Enter your manager name

David Lee

Enter your department name

Development

* Urgency

Medium

* Select License Type

LICENSED

Short description:

Description:

Work notes:

Figure 9: Software Request – Sent Email Evidence

Workflow Studio | SIRA Software Installation Workflow Studio

Test Run: Completed | Open Flow | Open context record

ACTORS	Flow Logic	Completed	Evaluated - True	Evaluated - False	Not Run	Progress
1. Get Catalog Variables from Request_Licensed_Software	Core Action	Completed				100%
2. IF "1 - Get Catalog Variables from Request_Licensed_Software" is High	Flow Logic		Evaluated - True	Evaluated - False		0%
3. Update Record	Core Action	Not Run				
4. Else	Flow Logic	Completed				100%
5. IF "1 - Get Catalog Variables from Request_Licensed_Software" is Medium	Flow Logic		Evaluated - True	Evaluated - False		0%
6. Update Record	Core Action	Completed				100%
7. Else	Flow Logic	Not Run				
8. IF "1 - Get Catalog Variables from Request_Licensed_Software" is Low	Flow Logic	Not Run				
9. Update Record	Core Action	Not Run				
10. Lock Up Records	Core Action	Completed				100%
11. For Each Item in "1 - Software License Keys" (1 of 1)	Flow Logic	Completed				100%
12. Set Flow Variables	Flow Logic	Completed				100%
13. Ask For Approval	Core Action	Completed				100%
14. IF Manager Approval Decision	Flow Logic	Completed				100%
15. IF Risk Level Decision	Flow Logic	Completed				100%
16. Ask For Approval	Core Action	Completed				100%
17. IF License Type Decision	Flow Logic	Completed				100%
18. Ask For Approval	Core Action	Completed				100%
19. IF Software Availability Decision	Flow Logic	Completed				100%
20. IF Software Asset Approval Decision	Flow Logic	Completed				100%
21. Create Catalog Task	Core Action	Completed				100%
22. Update Record	Core Action	Completed				100%
23. Send Email	Core Action	Completed				100%

Figure 10: Software Request – Email Record / Details

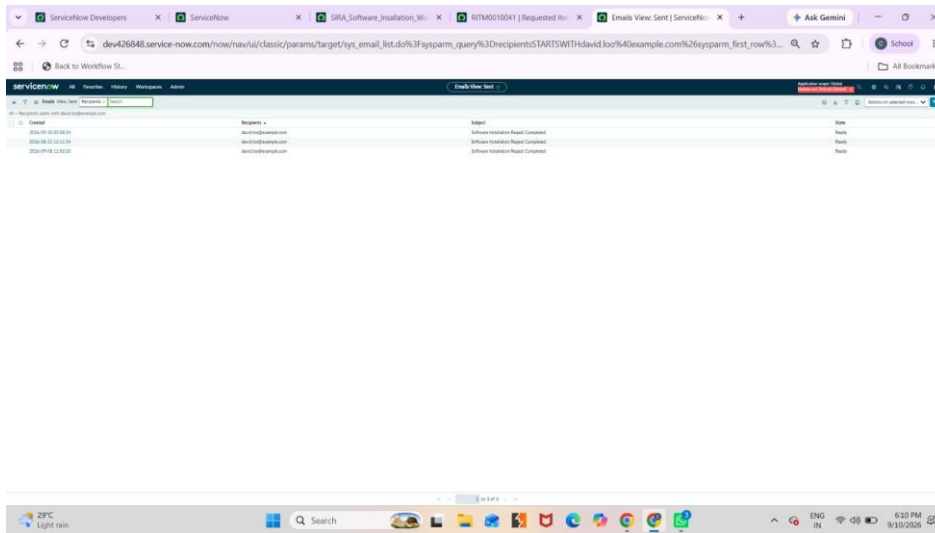


Figure 11: Software Request – Email Preview

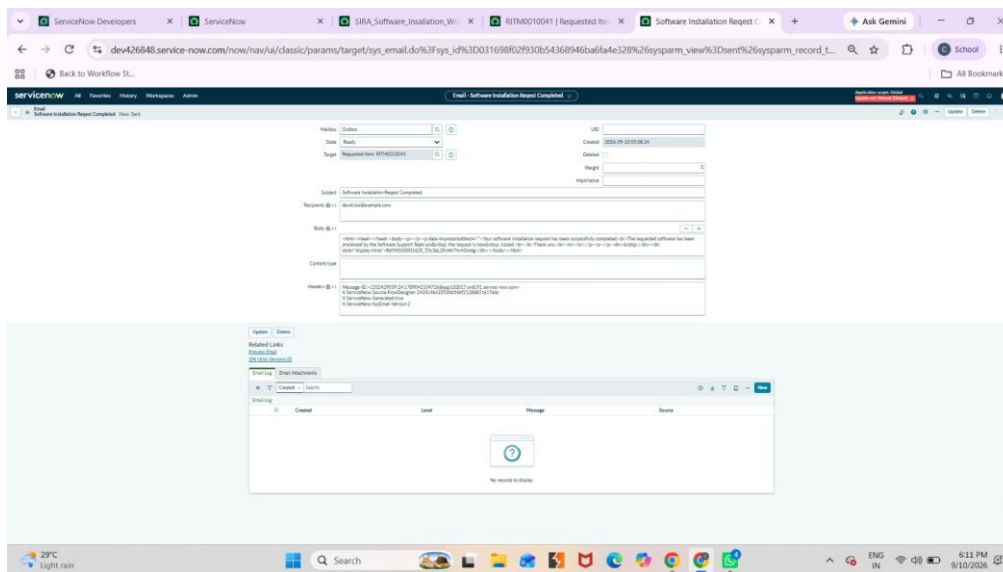


Figure 12: Software Request – High Priority SLA Configuration

The implementation also includes **SLA management, automated notifications, role-based security, fulfillment tracking, dashboard reporting, and Update Set deployment**. This reduces manual effort, improves request visibility, ensures proper approval before software installation, and supports better control over licensed software usage.

Overall, the project demonstrates how ServiceNow can be used to create a **standardized, traceable, secure, and efficient end-to-end software installation process**. The completed solution improves the experience for employees while helping IT and software support teams manage licensed software requests effectively.

Final Project Flow

Employee Request → Service Catalog → REQ/RITM → License Validation → Approval → Software Asset Approval → Catalog Task → Software Installation → Fulfillment → Notification → Dashboard → Closure